

LOUIS MARTIN

PROFESSIONAL AUDIO

K-SERIES

PROFESSIONAL COLUMN LOUDSPEAKER

USER MANUAL

Models Covered:

K302 · K304 · K306 · K308 · K312

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1. Product Overview

The **K-Series Column Loudspeaker** is a high-fidelity professional audio solution engineered for environments where speech intelligibility is paramount. Crafted from a robust **aluminium alloy cabinet**, it combines lightweight construction with excellent transient response, ensuring both durability and acoustic precision.

Designed specifically for acoustically challenging spaces — such as airports, churches, railway stations, and conference venues — the K-Series delivers uncompromising clarity in every environment. Each model is equipped with **3" Faital Pro full-range drivers**, offering consistent, reliable performance across the entire lineup.

Key Highlights

- Slim aluminium alloy cabinet — durable and lightweight
- 3" full-range Faital Pro driver(s) for consistent fidelity
- Easy setup with multiple mounting configuration options
- 90° horizontal dispersion with variable vertical coverage
- Passive full-range design — no active crossover required
- Nominal impedance: 8Ω (typical) — compatible with standard amplifiers
- Frequency response: 150 Hz – 18 kHz (–10 dB)

Design Philosophy

The K-Series is purpose-built for voice reproduction in reverberant spaces. The narrow vertical dispersion pattern of larger models (as low as 20°) concentrates acoustic energy toward the audience, minimising ceiling and floor reflections that degrade speech intelligibility.

2. Model Lineup & Images

The K-Series comprises five models, each progressively larger with more drivers, higher power handling, and tighter vertical dispersion. Select the model that best matches your venue coverage requirements.

				
K302	K304	K306	K308	K312
2 × 3" Drivers	4 × 3" Drivers	6 × 3" Drivers	8 × 3" Drivers	12 × 3" Drivers
114 × 101 × 110 mm	439 × 101 × 110 mm	649 × 101 × 110 mm	859 × 101 × 110 mm	1120 × 101 × 110 mm

Figure 1 – K-Series model lineup (left to right: K302, K304, K306, K308, K312). Heights shown proportionally.

3. Technical Specifications

All models share the same aluminium alloy cabinet construction and 8Ω nominal impedance. Performance scales with the number of drivers installed.

Parameter	K302	K304	K306	K308	K312
Type	Full Range Column Speaker (all models)				
Freq. Response (-10 dB)	150 Hz – 18 kHz (all models)				
Drivers	2 × 3" Full Range	4 × 3" Full Range	6 × 3" Full Range	8 × 3" Full Range	12 × 3" Full Range
RMS Power	40 W	80 W	120 W	160 W	240 W
Program Power	80 W	160 W	240 W	320 W	480 W
Peak Power	160 W	320 W	480 W	640 W	960 W
Sensitivity (1W/1m)	91 dB	93 dB	95 dB	97 dB	99 dB
Max SPL Cont. (1m)	110 dB	115 dB	118 dB	121 dB	125 dB
Max SPL Peak (1m)	113 dB	118 dB	121 dB	124 dB	128 dB
Nominal Impedance	8 Ω (typical — all models)				
Crossover	Passive Full Range (all models)				
Dispersion (H × V)	90° × 60°	90° × 50°	90° × 40°	90° × 30°	90° × 20°
Cabinet Material	Aluminium Alloy (all models)				

Table 1 – Complete technical specifications for the K-Series lineup.

4. Dimensions & Physical Data

All K-Series enclosures share identical width (101 mm) and depth (110 mm). Only the height varies across models. The table below lists the complete physical data including weight.

Model	Height (mm)	Width (mm)	Depth (mm)	Weight (kg)	Weight (lbs)
K302	114	101	110	2.2	4.85
K304	439	101	110	4.1	9.03
K306	649	101	110	5.9	13.0
K308	859	101	110	7.8	17.2
K312	1120	101	110	10.6	23.3

Table 2 – Physical dimensions and weight for all K-Series models.

Dimensional Reference

The diagrams below show each model to relative scale. The width (101 mm) and depth (110 mm) are identical across the entire lineup.

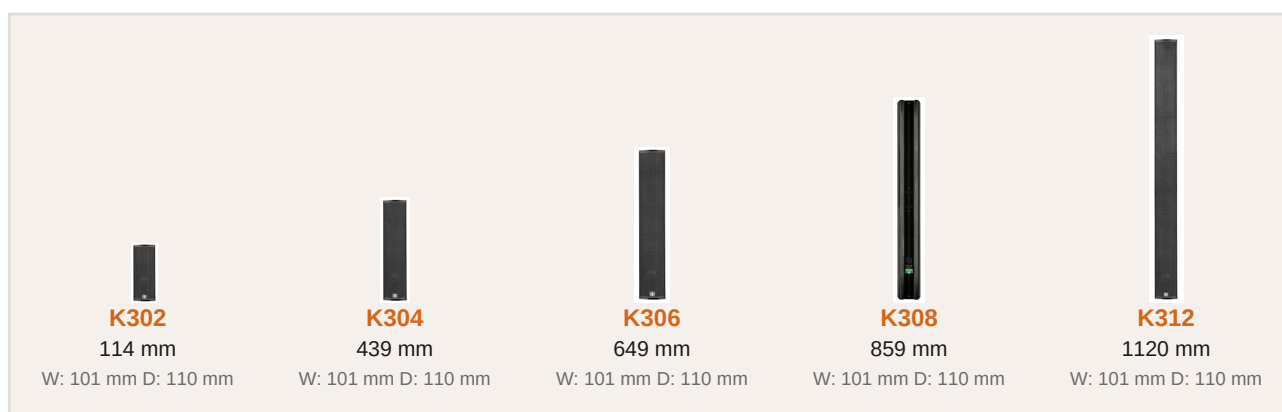


Figure 2 – K-Series models shown proportionally to relative height. All models: 101 mm wide × 110 mm deep.

5. Installation Guide

5.1 Pre-Installation Checklist

- Verify the amplifier output impedance matches the speaker nominal 8 Ohm impedance.
- Ensure the amplifier RMS power does not exceed the model rated RMS power limit.
- Check that the mounting surface or flying hardware is rated for at least 4× the speaker weight.
- Inspect all cables and connectors for damage before connection.
- Confirm the controller is programmed by an authorised Louis Martin technician.

5.2 Mounting Options

The K-Series has been designed for easy setup across multiple mounting configurations. Choose the configuration appropriate for your venue.

Mounting Type	Suitable Models	Notes
Wall bracket (fixed)	All models	Use supplied bracket; ensure wall anchor rated for 4× speaker weight
Ceiling / flying rigging	K306, K308, K312	Use certified flying hardware only; inspect regularly
Floor stand	K302, K304	Suitable for temporary or portable installations
Truss / line array bar	All models	Use appropriate clamps; confirm load rating with supplier

Table 3 – Recommended mounting configurations.

■ **IMPORTANT:** When flying speakers, appropriate certified hardware must be employed. All flying apparatus must be regularly inspected for any visual or mechanical failure. Never suspend speakers without verified load-rated hardware.

5.3 Aiming & Coverage

The K-Series maintains a constant 90° horizontal dispersion across all models. Vertical coverage narrows as model size increases, concentrating energy toward the listening area and reducing reflections from ceiling and floor surfaces.

Model	Horizontal Dispersion	Vertical Dispersion	Typical Application
K302	90°	60°	Small rooms, near-field reinforcement
K304	90°	50°	Medium rooms, classrooms
K306	90°	40°	Conference rooms, lecture theatres
K308	90°	30°	Large halls, churches
K312	90°	20°	Airports, railway stations, large venues

Table 4 – Dispersion characteristics and typical usage by model.

6. System Applications

Public Address Systems

The K-Series provides wide horizontal coverage and intelligible reproduction for general PA purposes. The narrow vertical beam of K308/K312 is ideal for long throw in open concourses.

Houses of Worship

Churches and places of worship benefit from the tight vertical dispersion of K306–K312, directing audio to pews while minimising reflections from high ceilings and hard walls.

Airports & Railway Stations

K308 and K312 excel in transit environments where long reverberation times demand precise directional control. Their high maximum SPL ensures announcements are heard even in noisy concourse environments.

Lecture Theatres & Classrooms

K304 and K306 offer the ideal balance of power and coverage for educational environments, delivering clear voice reproduction without excessive coloration.

Conference Rooms

K302 and K304 are well-suited for boardrooms and mid-size meeting spaces, providing natural speech reproduction at modest SPL requirements.

7. Connection & Wiring

The K-Series speakers are passive full-range loudspeakers. They require an external power amplifier. Always use the **Louis Martin digital controller** programmed by an authorised technician. Operating the system without the correct controller and program will void the warranty.

7.1 Amplifier Requirements

Parameter	Requirement
Output Impedance	8Ω (matched to speaker nominal impedance)
Minimum RMS Power	Equal to or greater than speaker RMS rating
Maximum RMS Power	Do not exceed the speaker RMS power rating
Signal Processing	Louis Martin digital controller (authorised setup only)

Table 5 – Amplifier and signal chain requirements.

7.2 Cable Recommendations

- Use oxygen-free copper (OFC) speaker cable for best performance.
- Minimum recommended cable gauge: 2.5 mm² (14 AWG) for runs up to 10 m.
- For runs exceeding 10 m, increase to 4 mm² (12 AWG) to minimise resistance losses.
- Ensure all connections are secure and free of corrosion.
- Label all cables clearly at both ends for ease of maintenance.

■ **IMPORTANT:** The Louis Martin digital controller must only be programmed by an authorised technician. Using the system without the correct controller configuration will void your warranty.

8. Safety Instructions

■ Flying & Rigging

When suspending speakers, appropriate certified hardware must be used for safely rigging the enclosures. All speaker hardware and flying apparatus must be regularly inspected for any visual or mechanical failure.

■ Audience Distance

Leave adequate distance between speakers and the audience. Refer to your local authority's Health and Safety guidance when operating loudspeaker systems.

■ Controller Use

The Louis Martin digital controller must only be programmed by an authorised technician. Do not operate Louis Martin loudspeaker systems without the correct controller and program loaded. Warranty is void if system failure results from incorrect controller use.

■ Environmental Protection

Protect speakers and all associated electronics from exposure to humidity, water, or rain. These products are not rated for outdoor installation unless specifically noted.

■ Hearing Safety

Prolonged exposure to high sound pressure levels can cause permanent hearing damage. Always observe local regulations regarding maximum SPL levels in public venues.

9. Warranty Information

Louis Martin loudspeakers and associated electronics are covered against defects in workmanship or materials for a period of **two (2) years** from the original date of purchase.

9.1 What is Covered

- Manufacturing defects in materials or workmanship
- Component failure under normal operating conditions
- Faulty driver or cabinet finish defects

9.2 What is NOT Covered

- Damage resulting from user abuse, accident, or negligence
- Modifications or repairs not authorised by Louis Martin
- Damage caused by use with incorrect controller settings
- Cosmetic damage due to improper storage or handling
- Damage from exposure to water, humidity, or extreme temperatures

9.3 Warranty Claim Procedure

1. Contact your authorised Louis Martin distributor to initiate a warranty claim.
2. Adequately package the defective item to prevent transit damage.
3. Dispatch the item pre-paid to the authorised distributor/service centre.
4. Louis Martin will repair or replace the defective item at no charge for materials or labour.
5. Repaired/replaced items are covered for the remainder of the original warranty period.

10. Contact & Support

For technical support, warranty claims, authorised service centres, or product enquiries, please contact Louis Martin Audio through the channels below.

Contact Type	Details
Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Warranty Claims	Contact your authorised Louis Martin distributor/service centre
Technical Support	Authorised technicians only — contact via email for referral

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Specifications are subject to change without notice. Always refer to the latest documentation available at www.louismartinaudio.it for the most current information. This manual is intended for professional installation personnel only.